

Subject Description Form

Subject Code	ME34005
Subject Title	Fundamentals of Thermal Science and Engineering
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AP10005 Physics I AMA2111 Mathematics I
Objectives	Thermal Science and Engineering (TSE) is a branch of science and engineering, covering topics in thermodynamics and heat transfer. These topics form the basic foundations that govern processes in engineering applications. This course is an introduction to TSE and how the principles can be applied to understand/design thermal processes. The specific course objectives are: (1) understand and apply principles of thermodynamics and heat transfer to engineering applications, and (2) form a foundation for subsequent studies in engineering thermodynamics, heat transfer, building services, material science, marine engineering, environmental engineering, power engineering, energy conversion, energy system and other areas.
Intended Learning Outcomes	a) Understand the fundamental principles of engineering thermodynamics and heat transfer. b) Formulate and solve thermal problems. c) Analyze and interpret data in thermal science and engineering. d) Present effectively in completing written reports.
Subject Synopsis/ Indicative Syllabus	Basic Concepts: Thermodynamics and applications; Systems; Properties; Extensive and intensive properties; Measured properties; Energy; Equilibrium; Zeroth law of thermodynamics; State and state principle; Process and cycle; Quasiequilibrium process. Properties of pure substances: Pure substances; Phase-change process; Property diagrams; Critical point and critical properties; Enthalpy; Property tables; Reference state; State equations for vapor phase; Ideal gases. Heat and work: Forms of energy; Energy transfer by heat; Adiabatic processes; Energy transfer by work; Mechanical forms of work; Moving boundary work; Polytropic processes. Energy analysis of closed systems: Specific heats; Internal energy; Enthalpy; First law of thermodynamics; Energy balance. Mass and energy analysis of open systems: Control volumes; Conservation of mass; Conservation of energy; Flow work and total energy of a flowing fluid; Steady-state and unsteady-flow processes; Applications in nozzles, diffusers, turbines, compressors, valves, mixing chambers and heat exchangers. Second law of thermodynamics: Thermal reservoirs; Power cycles; Efficiency; Refrigeration and heat pump cycles; Coefficients of performance; The Kelvin-Planck statement; The Clausius statement; Reversible and irreversible processes; Carnot principle; Carnot cycle; Thermodynamic temperature scale; Maximum

	performance measures for cycles; Quality of energy. Entropy: Clausius inequality; Entropy; Thermal entropy flux; Entropy production; Entropy of substances; Entropy changes; Entropy balance for closed systems; Principle of entropy increase; Entropy rate balance for open systems. Heat conduction and convection: Heat; Conduction; Forced convection; Natural convection; Fourier law of heat conduction; Thermal conductivity; Heat conduction equation; Newton law of cooling; Convective heat transfer coefficient; Nondimensional numbers; Empirical relations. Heat exchangers: Double-pipe heat exchangers; Shell-and-tube heat exchangers; Cross-flow heat exchangers; Overall heat-transfer coefficient; Log mean temperature difference; Effectiveness-NTU method; Design and applications.																																		
Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge and application in relation to thermodynamics and heat transfer (outcomes a-d). Tutorials are used to illustrate the application of fundamental knowledge to practical situations (outcomes a-d). <table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>Lecture</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Tutorial</td><td>√</td><td>√</td><td>√</td><td>√</td></tr></table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	Lecture	√	√	√	√	Tutorial	√	√	√	√															
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Examination</td><td>60%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tests</td><td>20%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>3. Homework</td><td>20%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100%</td><td colspan="4"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: 0.6 × End of Subject Examination + 0.4 × Continuous Assessment Examination is adopted to assess students on the overall understanding and the ability of applying the concepts and fundamental principles. It is supplemented by the tests (usually two) and homework (usually five) which provide timely feedbacks to both lecturers and students on various topics of the syllabus.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				a	b	c	d	1. Examination	60%	√	√	√	√	2. Tests	20%	√	√	√	√	3. Homework	20%	√	√	√	√	Total	100%				
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Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial	6 Hrs.
	Other student study effort:	
	▪ Course work	41 Hrs.
	▪ Self-study	40 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	1. Y.A. Cengel, M. A. Boles, M. Kanoglu. Thermodynamics: An Engineering Approach, McGraw Hill, 9th edition, in SI units. 2. R.E. Sonntag, C. Borgnakke and G.J.V. Wylen, Fundamentals of Thermodynamics, John Wiley and Son, 10th edition. 3. T.D. Eastop and A. McConkey, Applied Thermodynamics for Engineering Technologists, Pearson, 5th edition. 4. K. Wark, and D. Richards, Thermodynamics, McGraw-Hill, 6th edition. 5. J. P. Holman, Heat Transfer, McGraw Hill, 10th edition. 6. Y. A. Cengel, Heat transfer: A Practical Approach, McGraw Hill, 2nd edition. 7. K.D. Hagen, Heat Transfer with Applications, Prentice Hall, 1999. 8. T.L. Bergman, A.S. Lavine, F.D. Incropera, and D.P. Dewitt, Introduction to Heat Transfer, Wiley, 6th edition.	
Develop Date	September 2025	